ZCAS University

CIT3641 – ADVANCED DATABASES

DEFERRED END OF SEMESTER EXAMINATION

19th JUNE 2025

09:00 HRS – 12:00HRS

TIME ALLOWED: **WRITING – THREE HOURS**

READING TIME: 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer any **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Numbering of your answers in your booklet should be the same as that in the question paper. Answers that are not numbered will **NOT be marked**.
5. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

Normalization is one of the database design techniques. Its main goal is to reduce data redundancy (avoid storing the same data multiple times), provide for consistency, prevent anomalies (insert, update and delete anomalies) and improve on data integrity. There are seven (7) levels of normalization. Each Level / Normal Form of Normalization:

- addresses a specific problem in database design process,
- builds upon the previous level, and
- contributes toward the main goal of Normalization.

Requirements:

i) You are required to explain the rules or conditions which should be satisfied / applicable at each of the seven (7) levels / Normal Forms of Normalization. **(35 Marks)**

ii) Using an appropriate database table for illustration purposes, explain the meaning of an insert anomaly, an update anomaly and a delete anomaly. **(5 Marks)**

(Total: 40 Marks)

SECTION B

Question 2

In database management systems, ACID (Atomicity, Consistency, Isolation, and Durability) is a set of properties that ensure transactions are executed reliably, preventing issues such as data corruption, inconsistency, or loss. These properties are especially crucial in multi-user environments where transactions occur concurrently.

A transaction is a sequence of one or more database operations that are treated as a single unit of work. ACID properties help ensure that these transactions maintain data integrity, even in cases of failures such as power outages, system crashes, or concurrent transaction conflicts.

Requirement:

Using an appropriate example for each property, explain the specific operations associated with each of the four (4) ACID properties.

NOTE: Each property should be explained separately with the property name clearly indicated.

(Total: 20 Marks)

Question 3

i) A local hobby club organises various workshops throughout the year, covering different activities such as painting, photography, and cooking. Each workshop can be attended by many members of the club, and each member can sign up for multiple workshops based on their interests.

Requirement:

Draw an Entity Relationship Diagram to represent the scenario described above. You need to identify entities, relationships, cardinality, label all the primary keys and foreign keys. Clearly state which entity the keys belong to. **(10 Marks)**

ii) A multivalued attribute is an attribute that can be captured more than once in an entity. One simple example is that of multiple mobile phone numbers belonging to one student.

Requirements:

a) Explain how students and their mobile phone numbers (zero, one or many) may be represented or structured in a small database table(s). **(5 Marks)**

b) State and explain the actual Structured Query Language (SQL) commands which may be used to create the appropriate table(s) identified in ii) (a) above. **(5 Marks)**

(Total: 20 Marks)

Question 4

i) The Structured Query Language (SQL) Statement below creates an Employee_Salary table designed to track salary changes for the employees. Each change in the salary for an employee result into an INSERT of a new record into the table. It also means the details for the previous salary for the same employee remain unchanged / intact in the table and continues to be available for queries.

```
CREATE TABLE Employee_Salary (
Employee_ID INT,
Salary DECIMAL(10,2),
Start_Date DATE,
End_Date DATE,
PRIMARY KEY (Employee_ID, Start_Date));
```

Requirement:

Explain why the use of a composite primary key is necessary in the table created above.

(5 Marks)

ii) With the consideration of the Employee_Salary table created in (i) above, study the INSERT SQL Statement indicated below.

```
INSERT INTO Employee_Salary
    (Start_Date, Salary, Employee_ID, End_Date)
VALUES
    ('2023-01-01', 50000, 101, '2023-06-30'),
    ('2023-07-01', 55000, 101, '2023-12-31'),
    ('2023-02-01', 60000, 102, '2023-12-31');
```

Requirement:

State as to whether the INSERT Statement provided will successfully execute or will fail. Explain to justify your answer (Why it will successfully execute or why it will fail).

(5 Marks)

Let us assume that the table (Employee_Salary) created in (i) above now has the four (4) records indicated in the table below. Use the table to answer Part (iii) and Part (iv) below:

Employee_Salary

Employee_ID	Salary	Start_Date	End_Date
101	50000	2023-01-01	2023-06-30
102	60000	2023-02-01	2023-12-31
101	55000	2023-07-01	2023-12-31
103	40000	2023-08-01	2024-05-31

iii) **Requirement:**

Using the Employee_Salary table above, which SQL Statement can be used to answer the query below:

What was Employee 101's Salary in March 2023?

Provide an output for your proposed SQL Statement. **(5 Marks)**

iv) **Requirement:**

Using the Employee_Salary table above, which SQL Statement can be used to answer the query below:

Display Employee_IDs of Employees whose Salaries changed in 2023.

Provide an output for your proposed SQL Statement. **(5 Marks)**

(Total: 20 Marks)

Question 5

i) The following are concepts or terminologies that are associated with Distributed Database Systems. In one sentence or at most one paragraph, you are required to provide a brief explanation of EACH concept or terminology:

a) Fragmentation,

b) Replication,

c) Homogeneous Distributed Database System,

d) Heterogeneous Distributed Database Systems. **(8 Marks)**

ii) The following are the four (4) key features of a Data Warehouse. In one paragraph, you are required to provide a brief writeup to explain EACH of the four (4) features indicated below:

a) Subject-oriented,

b) Integrated,

c) Time-variant,

d) Non-volatile. **(12 Marks)**

NOTE: Each of the concepts / features, in (i) and (ii) above should be correctly numbered and clearly labelled.

(Total: 20 Marks)

END OF EXAMINATION